

KVITFJELL, Norway

WMO: 013750

Lat: 61.465N

Lon: 10.127E

Elev: 3373

StdP: 12.99

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	2.7	-7.1	4.2	4.6	-3.1	5.3	3.8	40.5	23.5	34.9	24.6	5.0	160	0.873

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	67.9	54.6	64.5	53.5	61.0	52.0	57.1	64.0	55.2	60.6	53.5	58.6	9.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.5	71.7	58.5	53.0	67.6	57.0	51.2	63.3	55.2	25.9	64.3	24.6	60.6	23.6	58.8	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
33.0	28.6	24.9	-2.5	71.6	4.8	4.5	-5.9	74.8	-8.7	77.4	-11.4	80.0	-14.9	83.3
			-3.0	58.9	4.6	2.6	-6.4	60.8	-9.1	62.3	-11.7	63.7	-15.0	65.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	18.0	19.1	23.1	31.1	39.6	47.4	52.7	49.1	42.5	33.3	25.2	21.5
	DBStd	13.75	7.50	8.19	7.69	6.81	8.07	6.67	6.12	4.53	5.46	5.90	7.21	7.77
	HDD50	6211	992	866	834	567	339	128	35	73	233	518	743	883
	HDD65	11453	1457	1286	1299	1017	787	529	384	494	676	983	1193	1348
	CDD50	236	0	0	0	0	17	49	119	44	7	0	0	0
	CDD65	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.4	10.7	10.8	12.2	11.5	11.4	11.6	10.8	10.4	12.1	12.2	11.5	11.2
Precipitation	PrecAvg	27.3	1.9	1.3	1.2	1.4	2.4	2.9	3.7	3.8	2.3	2.5	2.4	1.6
	PrecMax	41.1	3.9	2.9	3.0	3.4	5.7	5.8	6.7	6.3	4.2	8.0	9.3	3.2
	PrecMin	20.2	0.2	0.4	0.0	0.0	0.2	0.1	1.0	0.6	0.7	0.4	0.4	0.1
	PrecStd	4.1	0.9	0.6	0.8	0.7	1.3	1.4	1.3	1.5	0.9	1.6	1.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	39.4	44.9	52.0	66.3	70.8	73.2	64.2	59.4	51.8	43.9	39.6
		MCWB	33.5	35.0	36.6	42.8	50.6	53.0	58.1	55.5	51.4	47.8	38.7	36.5
	2%	DB	34.1	35.7	40.4	47.5	61.0	66.6	69.5	61.3	55.7	46.3	40.2	36.6
		MCWB	31.1	31.2	34.7	39.5	49.4	51.9	57.3	53.6	49.4	41.9	37.4	33.5
	5%	DB	30.9	32.8	37.3	44.3	56.1	61.9	66.2	58.6	52.7	43.5	37.3	34.3
		MCWB	29.2	29.6	32.7	37.7	46.4	50.3	55.1	52.1	48.0	39.9	35.7	31.4
	10%	DB	27.9	29.7	33.6	41.3	51.8	57.7	63.0	55.9	50.1	41.4	34.6	31.6
		MCWB	26.6	27.5	30.1	35.4	44.6	48.2	53.9	51.6	46.3	38.9	33.1	29.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	35.6	38.3	43.9	53.3	55.3	61.2	57.5	53.8	49.9	40.5	36.8
		MCDB	36.4	38.4	42.9	50.0	61.7	62.1	69.6	61.4	56.7	51.2	42.1	38.7
	2%	WB	31.5	32.2	35.0	40.4	50.4	53.5	58.6	55.5	50.9	43.6	37.5	34.1
		MCDB	33.5	34.7	39.6	45.4	58.1	61.7	66.9	58.4	53.7	45.1	39.4	36.1
	5%	WB	29.3	29.8	32.8	38.5	47.8	51.9	56.6	54.1	48.8	41.3	35.5	31.8
		MCDB	30.8	32.1	36.8	43.5	54.8	59.3	64.0	56.9	51.4	42.6	37.2	33.5
	10%	WB	26.7	27.6	30.5	36.0	45.2	49.9	54.7	52.2	47.0	39.3	33.3	29.7
		MCDB	27.9	29.7	33.3	40.7	50.6	57.2	61.0	55.1	49.6	40.9	34.5	31.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	8.4	9.4	11.1	12.5	11.6	9.7	8.9	7.0	6.4	6.8
		MCDBR	9.1	9.8	11.1	12.6	15.9	16.5	14.8	12.5	12.0	10.0	9.0	10.1
		MCWBR	8.0	8.0	7.8	7.8	8.4	8.4	7.4	7.2	7.7	7.6	7.6	9.1
	5% WB	MCDBR	8.8	9.2	10.6	11.5	14.4	13.7	13.0	10.1	10.2	8.8	8.8	9.6
		MCWBR	8.0	8.0	7.9	7.5	8.5	8.0	7.4	6.9	7.4	8.0	7.7	9.2
Clear-Sky Solar Irradiance	taub	0.191	0.236	0.265	0.296	0.307	0.305	0.317	0.310	0.282	0.265	0.204	0.157	
	taud	2.141	2.290	2.289	2.306	2.480	2.544	2.510	2.522	2.587	2.520	2.207	1.984	
	Ebn at Noon	186	247	275	286	292	294	286	280	272	236	176	139	
	Edh at Noon	10	19	27	33	30	29	29	26	21	15	10	7	
All-Sky Solar Radiation	RadAvg	92	306	802	1366	1539	1638	1498	1160	748	338	112	51	
	RadStd	15	59	79	135	153	158	181	97	86	53	21	4	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page